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Section-B

Course:B.tech Cse Core

Assignment 3 :

❖ Aim

To design and develop a full-stack Quiz Master web application using HTML, CSS, Python, and Flask framework, enabling users to attempt multiple-choice questions and view their scores with feedback.

❖ Tasks

- Set up a Flask project and application
- Create quiz questions and store them (static or database)
- Design HTML templates for Home, Quiz, and Result pages
- Implement backend logic using Flask views
- Handle form submission and user responses
- Calculate quiz score based on answers
- Display result and feedback to the user
- Apply CSS styling for better user interface

❖ Technologies Used

- **Frontend:** HTML, CSS
- **Backend:** Python, Flask
- **Framework:** Flask Web Framework
- **Tools:** VS Code / PyCharm, Web Browser
- **Version Control (Optional):** Git

❖ Code :

〉HTML CODE :

❖ Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Quiz Master - Home</title>
<link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
<div class="page-wrapper">
<header class="site-header">
<div class="logo">🎯 Quiz Master</div>
<nav class="nav-links">
<span>Web Programming Lab</span>
</nav>
</header>

<main class="home-main">
<div class="hero-card">
<div class="hero-icon">□</div>
<h1 class="hero-title">Welcome to <span class="accent">Quiz Master</span></h1>
<p class="hero-subtitle">Test your knowledge in Web Development, Python &
Flask</p>

<div class="info-grid">
<div class="info-box">
<span class="info-icon">📝 </span>
<span class="info-label">10 Questions</span>
</div>
<div class="info-box">
<span class="info-icon">⏱</span>
<span class="info-label">5 Min Timer</span>
</div>
<div class="info-box">
```

```
<span class="info-icon">🔀</span>
<span class="info-label">Shuffled</span>
</div>
<div class="info-box">
<span class="info-icon">➖</span>
<span class="info-label">-0.25 Marking</span>
</div>
</div>

<div class="rules-box">
<h3>📋 Rules</h3>
<ul>
<li>Each correct answer earns <strong>+1 mark</strong></li>
<li>Each wrong answer deducts <strong>-0.25 marks</strong></li>
<li>Skipped questions carry <strong>no penalty</strong></li>
<li>Questions and options are <strong>shuffled</strong> every attempt</li>
<li>Timer auto-submits at <strong>5 minutes</strong></li>
</ul>
</div>

<a href="/start" class="btn-start">
🚀 Start Quiz
</a>
</div>
</main>

<footer class="site-footer">
<p>Assignment 3 &mdash; Web Programming with Python and JavaScript Lab</p>
</footer>
</div>
</body>
</html>
```

❖ **Quiz.html**

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Quiz Master - Quiz</title>
<link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
<div class="page-wrapper">
<header class="site-header">
<div class="logo">🎯 Quiz Master</div>
<div class="timer-display" id="timer">
⏱ <span id="time-left">{{ '%02d:%02d' % (remaining // 60, remaining % 60)
  }}</span>
</div>
</header>

<main class="quiz-main">
<div class="quiz-progress-bar">
<div class="progress-fill" id="progress-fill" style="width: 0%"></div>
</div>
<div class="quiz-meta">
<span id="answered-count">0</span> / {{ questions|length }} answered
</div>

<form method="POST" action="/submit" id="quiz-form">
{% for q in questions %}
<div class="question-card" id="qcard-{{ loop.index }}">
<div class="question-header">
<span class="q-number">Q{{ loop.index }}</span>
<span class="q-status" id="status-{{ q.id }}">☐ Not answered</span>
</div>
<p class="question-text">{{ q.question }}</p>
<div class="options-grid">
{% for opt in q.options %}
<label class="option-label" for="q{{ q.id }}_{{ loop.index }}">
<input
type="radio"
name="q{{ q.id }}"
id="q{{ q.id }}_{{ loop.index }}"
value="{{ opt }}"
onchange="markAnswered({{ q.id }}, {{ loop.index0 }})"
>
```

```
<span class="option-text">{{ opt }}</span>
</label>
{% endfor %}
</div>
</div>
{% endfor %}

<div class="submit-section">
<button type="button" class="btn-submit" onclick="confirmSubmit()">
✅ Submit Quiz
</button>
</div>
</form>
</main>

<footer class="site-footer">
<p>Assignment 3 &mdash; Web Programming with Python and JavaScript
    Lab</p>
</footer>
</div>

<!-- Confirm Modal -->
<div class="modal-overlay" id="modal" style="display:none;">
<div class="modal-box">
<h2>Submit Quiz?</h2>
<p id="modal-msg">Are you sure you want to submit?</p>
<div class="modal-buttons">
<button class="btn-confirm" onclick="document.getElementById('quiz-
    form').submit()">Yes, Submit</button>
<button class="btn-cancel" onclick="closeModal()">Cancel</button>
</div>
</div>
</div>

<script>
// Timer
let remaining = {{ remaining }};
const totalTime = {{ timer_total }};
const timerEl = document.getElementById('time-left');
const timerDisplay = document.getElementById('timer');

function updateTimer() {
if (remaining <= 0) {
```

```
timerEl.textContent = "00:00";
alert("⏰ Time's up! Submitting your quiz...");
window.location.href = '/submit_timeout';
return;
}
remaining--;
const m = Math.floor(remaining / 60);
const s = remaining % 60;
timerEl.textContent = (m < 10 ? '0' : '') + m + ':' + (s < 10 ? '0' : '') + s;
if (remaining <= 60) {
timerDisplay.classList.add('timer-urgent');
} else {
timerDisplay.classList.remove('timer-urgent');
}
}
setInterval(updateTimer, 1000);

// Track answered questions
const answered = new Set();
const totalQ = {{ questions|length }};

function markAnswered(qid, optIdx) {
answered.add(qid);
const statusEl = document.getElementById('status-' + qid);
if (statusEl) statusEl.textContent = '✅ Answered';
updateProgress();
}

function updateProgress() {
const pct = (answered.size / totalQ) * 100;
document.getElementById('progress-fill').style.width = pct + '%';
document.getElementById('answered-count').textContent = answered.size;
}

function confirmSubmit() {
const unanswered = totalQ - answered.size;
const msgEl = document.getElementById('modal-msg');
if (unanswered > 0) {
msgEl.textContent = `You have ${unanswered} unanswered question(s). Submit
    anyway?`;
} else {
msgEl.textContent = "All questions answered. Ready to submit!";
}
```

```
document.getElementById('modal').style.display = 'flex';
}

function closeModal() {
document.getElementById('modal').style.display = 'none';
}
</script>
</body>
</html>
```

❖ **result.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Quiz Master - Results</title>
<link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
<div class="page-wrapper">
<header class="site-header">
<div class="logo">🎯 Quiz Master</div>
<nav class="nav-links"><span>Results</span></nav>
</header>

<main class="result-main">
<!-- Score Card -->
<div class="score-card">
<div class="grade-badge grade-{{ grade | replace('+','plus') }}">{{ grade }}</div>
<h1 class="score-title">Quiz Completed!</h1>
<p class="feedback-msg">{{ feedback }}</p>
```

```
<div class="score-stats">
<div class="stat-box correct-box">
<span class="stat-num">{{ score }}</span>
<span class="stat-label">Correct</span>
</div>
<div class="stat-box wrong-box">
<span class="stat-num">{{ (total - score - results | selectattr('skipped') | list | length) }}</span>
<span class="stat-label">Wrong</span>
</div>
<div class="stat-box skip-box">
<span class="stat-num">{{ results | selectattr('skipped') | list | length }}</span>
<span class="stat-label">Skipped</span>
</div>
<div class="stat-box final-box">
<span class="stat-num">{{ "%.2f" | format(final_score) }} / {{ total }}</span>
<span class="stat-label">Final Score</span>
</div>
</div>

<div class="extra-stats">
<span>⏱ Time Taken: <strong>{{ time_taken }}</strong></span>
<span>➖ Penalty: <strong>-{{ "%.2f" | format(negative) }}</strong></span>
</div>

<div class="result-actions">
<a href="/start" class="btn-restart">🔄 Restart Quiz</a>
<a href="/" class="btn-home">🏠 Home</a>
</div>
```

```
</div>

<!-- Detailed Review -->
<div class="review-section">
<h2>📖 Answer Review</h2>
{% for r in results %}
<div class="review-card {% if r.correct %}card-correct{% elif r.skipped %}card-skipped{% else %}card-wrong{% endif %}">
<div class="review-header">
<span class="review-qnum">Q{{ loop.index }}</span>
<span class="review-status">
{% if r.correct %}✅ Correct (+1)
{% elif r.skipped %}☐ Skipped (0)
{% else %}❌ Wrong (-0.25)
{% endif %}
</span>
</div>
<p class="review-question">{{ r.question }}</p>
<div class="review-options">
{% for opt in r.options %}
<div class="review-option
{% if opt == r.answer %}opt-correct{% endif %}
{% if opt == r.selected and not r.correct %}opt-wrong{% endif %}">
{% if opt == r.answer %}✅ {% elif opt == r.selected %}❌ {% else %}○ {% endif %}
{{ opt }}
</div>
{% endfor %}
</div>
{% if r.skipped %}
<p class="review-note">You skipped this question. Correct answer: <strong>{{ r.answer }}</strong></p>
```

```
    {% endif %}
    </div>
    {% endfor %}
    </div>
    </main>

    <footer class="site-footer">
    <p>Assignment 3 &mdash; Web Programming with Python and JavaScript Lab</p>
    </footer>
    </div>
    </body>
</html>
```

❖ **style.css**

```
/* ========== RESET & BASE ========== */
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }

:root {
--primary:    #1417bd;
--primary-dk: #6b65d3;
--accent:     #FF6584;
--success:    #2ECC71;
--danger:     #E74C3C;
--warn:       #F39C12;
--bg:         #F0F2FF;
--card:       #FFFFFF;
--text:       #2D2D3A;
--muted:      #7A7A99;
--border:     #E0E0F0;
--radius:     14px;
--shadow:     0 4px 24px rgba(108,99,255,0.10);
}

body {
font-family: 'Segoe UI', system-ui, sans-serif;
background: var(--bg);
color: var(--text);
min-height: 100vh;
```

```
}

a { text-decoration: none; color: inherit; }

/* ========== LAYOUT ========== */
.page-wrapper {
display: flex;
flex-direction: column;
min-height: 100vh;
}

/* ========== HEADER ========== */
.site-header {
background: var(--primary);
color: #fff;
padding: 0 2rem;
height: 60px;
display: flex;
align-items: center;
justify-content: space-between;
position: sticky;
top: 0;
z-index: 100;
box-shadow: 0 2px 12px rgba(108,99,255,0.25);
}

.logo {
font-size: 1.3rem;
font-weight: 700;
letter-spacing: 0.5px;
}

.nav-links { font-size: 0.9rem; opacity: 0.85; }

/* Timer */
.timer-display {
font-size: 1.1rem;
font-weight: 700;
background: rgba(255,255,255,0.2);
padding: 6px 16px;
border-radius: 30px;
transition: background 0.3s;
}
.timer-urgent {
background: var(--accent) !important;
animation: pulse 1s infinite;
}
```

```
@keyframes pulse {
0%, 100% { opacity: 1; }
50%      { opacity: 0.7; }
}

/* ========== FOOTER ========== */
.site-footer {
margin-top: auto;
text-align: center;
padding: 1rem;
font-size: 0.8rem;
color: var(--muted);
border-top: 1px solid var(--border);
}

/* ========== HOME PAGE ========== */
.home-main {
flex: 1;
display: flex;
align-items: center;
justify-content: center;
padding: 2rem 1rem;
}

.hero-card {
background: var(--card);
border-radius: var(--radius);
box-shadow: var(--shadow);
padding: 3rem 2.5rem;
max-width: 580px;
width: 100%;
text-align: center;
}

.hero-icon { font-size: 4rem; margin-bottom: 1rem; }

.hero-title {
font-size: 2rem;
font-weight: 800;
margin-bottom: 0.5rem;
}

.hero-title .accent { color: var(--primary); }

.hero-subtitle {
color: var(--muted);
margin-bottom: 2rem;
```

```
font-size: 1.05rem;
}

.info-grid {
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 0.75rem;
margin-bottom: 2rem;
}

.info-box {
background: var(--bg);
border-radius: 10px;
padding: 0.75rem 0.5rem;
display: flex;
flex-direction: column;
align-items: center;
gap: 4px;
}

.info-icon { font-size: 1.4rem; }
.info-label { font-size: 0.75rem; font-weight: 600; color: var(--muted); }

.rules-box {
background: #F7F7FF;
border: 1px solid var(--border);
border-radius: 10px;
padding: 1.25rem 1.5rem;
text-align: left;
margin-bottom: 2rem;
}

.rules-box h3 { margin-bottom: 0.75rem; font-size: 1rem; }

.rules-box ul {
list-style: none;
display: flex;
flex-direction: column;
gap: 0.4rem;
}

.rules-box ul li {
font-size: 0.9rem;
color: var(--muted);
padding-left: 1rem;
position: relative;
}
```

```
.rules-box ul li::before {
content: '›';
position: absolute;
left: 0;
color: var(--primary);
font-weight: 700;
}

.btn-start {
display: inline-block;
background: var(--primary);
color: #fff;
font-size: 1.1rem;
font-weight: 700;
padding: 0.9rem 2.5rem;
border-radius: 50px;
transition: background 0.2s, transform 0.15s;
box-shadow: 0 4px 16px rgba(108,99,255,0.35);
}
.btn-start:hover { background: var(--primary-dk); transform: translateY(-2px); }

/* ========== QUIZ PAGE ========== */
.quiz-main {
flex: 1;
max-width: 760px;
margin: 0 auto;
width: 100%;
padding: 1.5rem 1rem 3rem;
}

.quiz-progress-bar {
height: 6px;
background: var(--border);
border-radius: 99px;
margin-bottom: 0.5rem;
overflow: hidden;
}

.progress-fill {
height: 100%;
background: linear-gradient(90deg, var(--primary), var(--accent));
border-radius: 99px;
transition: width 0.4s ease;
}

.quiz-meta {
text-align: right;
```

```
font-size: 0.85rem;
color: var(--muted);
margin-bottom: 1.5rem;
}

.question-card {
background: var(--card);
border-radius: var(--radius);
box-shadow: var(--shadow);
padding: 1.5rem;
margin-bottom: 1.25rem;
border-left: 4px solid var(--primary);
transition: box-shadow 0.2s;
}
.question-card:hover { box-shadow: 0 6px 32px rgba(108,99,255,0.15); }

.question-header {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 0.75rem;
}

.q-number {
font-size: 0.78rem;
font-weight: 700;
text-transform: uppercase;
letter-spacing: 1px;
color: var(--primary);
background: #EEF0FF;
padding: 3px 10px;
border-radius: 20px;
}

.q-status {
font-size: 0.8rem;
color: var(--muted);
}

.question-text {
font-size: 1.05rem;
font-weight: 600;
line-height: 1.5;
margin-bottom: 1rem;
}

.options-grid {
```

```
display: flex;
flex-direction: column;
gap: 0.6rem;
}

.option-label {
display: flex;
align-items: center;
gap: 0.75rem;
padding: 0.7rem 1rem;
border-radius: 10px;
border: 2px solid var(--border);
cursor: pointer;
transition: border-color 0.2s, background 0.2s;
font-size: 0.95rem;
}
.option-label:hover { border-color: var(--primary); background: #F5F4FF; }

.option-label input[type="radio"] { accent-color: var(--primary); width: 17px; height:
17px; cursor: pointer; }

.option-label:has(input:checked) {
border-color: var(--primary);
background: #EEEEFF;
}

.submit-section {
text-align: center;
margin-top: 2rem;
}

.btn-submit {
background: var(--success);
color: #fff;
font-size: 1.1rem;
font-weight: 700;
padding: 0.9rem 2.5rem;
border: none;
border-radius: 50px;
cursor: pointer;
box-shadow: 0 4px 16px rgba(46,204,113,0.35);
transition: background 0.2s, transform 0.15s;
}
.btn-submit:hover { background: #27ae60; transform: translateY(-2px); }

/* Modal */
.modal-overlay {
```

```
position: fixed; inset: 0;
background: rgba(0,0,0,0.45);
display: flex;
align-items: center;
justify-content: center;
z-index: 999;
}

.modal-box {
background: var(--card);
border-radius: var(--radius);
padding: 2rem;
max-width: 400px;
width: 90%;
text-align: center;
box-shadow: 0 8px 40px rgba(0,0,0,0.2);
}

.modal-box h2 { font-size: 1.4rem; margin-bottom: 0.75rem; }
.modal-box p  { color: var(--muted); margin-bottom: 1.5rem; }

.modal-buttons { display: flex; gap: 1rem; justify-content: center; }

.btn-confirm {
background: var(--primary);
color: #fff;
padding: 0.65rem 1.5rem;
border: none;
border-radius: 30px;
font-weight: 700;
cursor: pointer;
font-size: 0.95rem;
}
.btn-cancel {
background: var(--border);
color: var(--text);
padding: 0.65rem 1.5rem;
border: none;
border-radius: 30px;
font-weight: 600;
cursor: pointer;
font-size: 0.95rem;
}
.btn-confirm:hover { background: var(--primary-dk); }
.btn-cancel:hover  { background: #D0D0E8; }

/* ========== RESULT PAGE ========== */
```

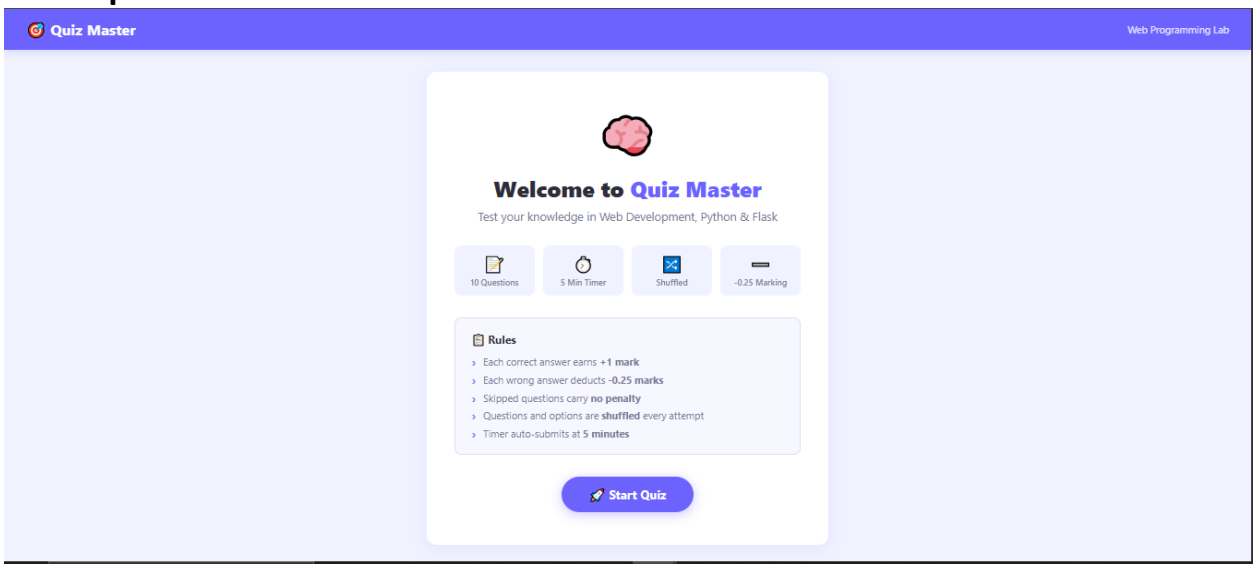
```
.result-main {
flex: 1;
max-width: 760px;
margin: 0 auto;
width: 100%;
padding: 1.5rem 1rem 3rem;
}

.score-card {
background: var(--card);
border-radius: var(--radius);
box-shadow: var(--shadow);
padding: 2.5rem 2rem;
text-align: center;
margin-bottom: 2rem;
}

.grade-badge {
display: inline-block;
font-size: 2.5rem;
font-weight: 900;
width: 80px; height: 80px;
line-height: 80px;
border-radius: 50%;
margin-bottom: 1rem;
color: #fff;
}
.grade-Aplus, .grade-A { background: var(--success); }
.grade-B             { background: var(--primary); }
.grade-C             { background: var(--warn); }
.grade-F, .grade---   { background: var(--danger); }

.score-title {
font-size: 1.8rem;
font-weight: 800;
margin-bottom: 0.5rem;
}

.feedback-msg {
font-size: 1.05rem;
color: var(--muted);
margin-bottom: 1.75rem;
}

.score-stats {
display: grid;
grid-template-columns: repeat(4, 1fr);
```

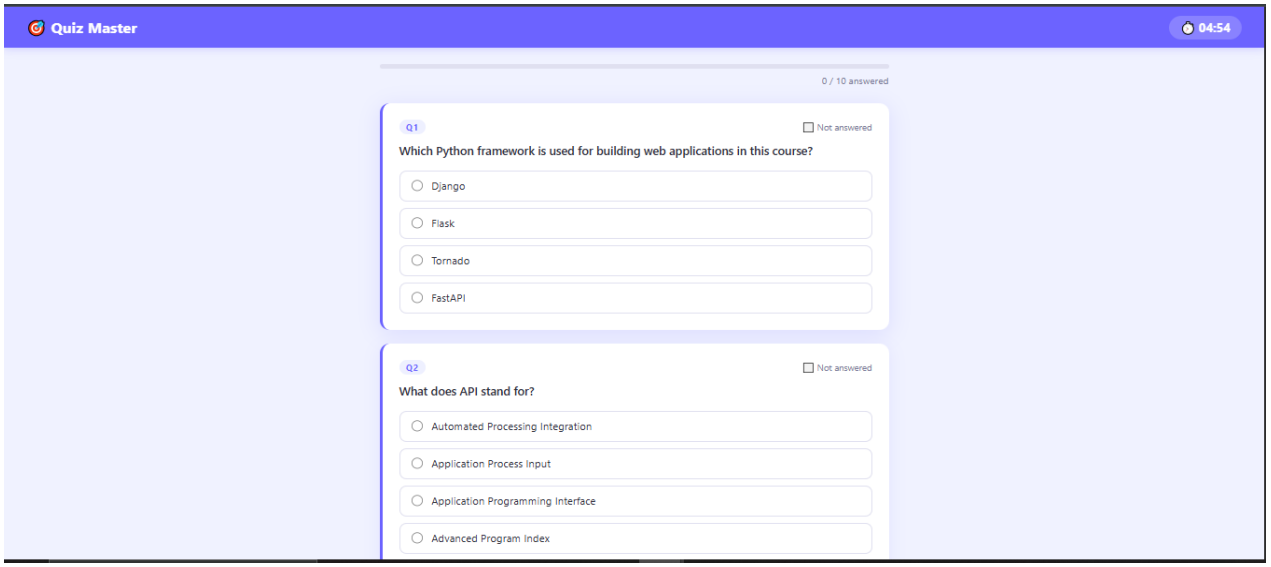
```
gap: 0.75rem;
margin-bottom: 1rem;
}

.stat-box {
border-radius: 12px;
padding: 1rem 0.5rem;
display: flex;
flex-direction: column;
align-items: center;
gap: 4px;
}

.stat-num { font-size: 1.4rem; font-weight: 800; }
.stat-label { font-size: 0.75rem; font-weight: 600; text-transform: uppercase; }

.correct-box { background: #D5F5E3; color: #1D8348; }
.wrong-box   { background: #FADBD8; color: #922B21; }
.skip-box    { background: #FEF9E7; color: #9A7D0A; }
.final-box   { background: #EAF2FF; color: #1A5276; }

.extra-stats {
display: flex;
justify-content: center;
gap: 2rem;
font-size: 0.9rem;
color: var(--muted);
margin-top: 0.75rem;
margin-bottom: 1.5rem;
}

.result-actions {
display: flex;
gap: 1rem;
justify-content: center;
}

.btn-restart, .btn-home {
padding: 0.75rem 1.75rem;
border-radius: 30px;
font-weight: 700;
font-size: 0.95rem;
transition: transform 0.15s;
}
.btn-restart { background: var(--primary); color: #fff; box-shadow: 0 4px 16px
rgba(108,99,255,0.3); }
.btn-home    { background: var(--border); color: var(--text); }
```

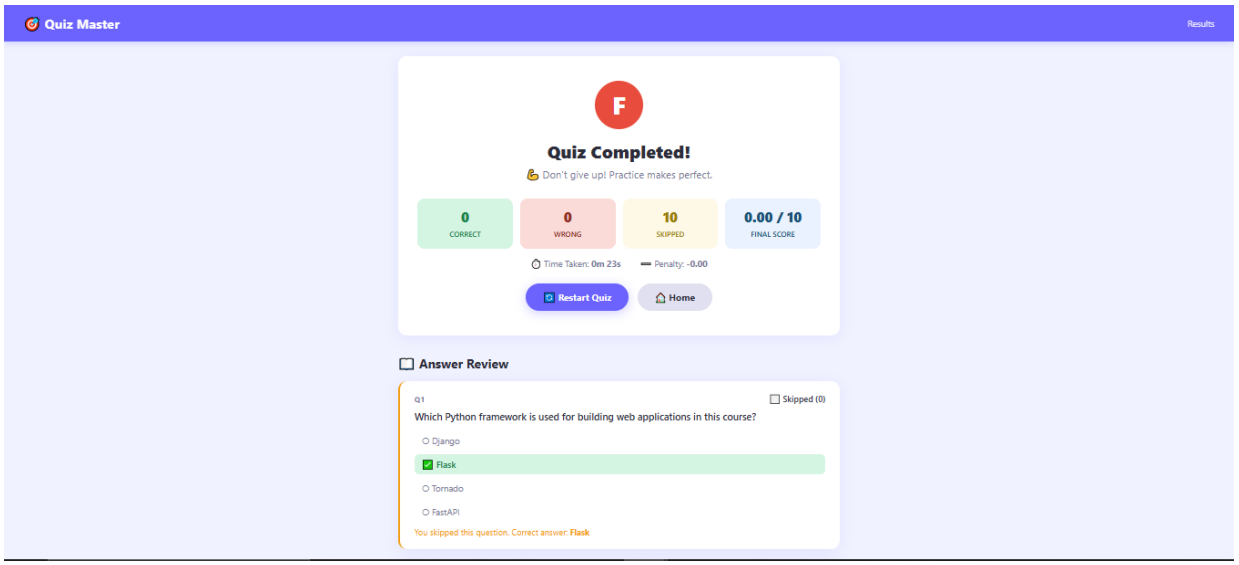
```
.btn-restart:hover, .btn-home:hover { transform: translateY(-2px); }

/* Review */
.review-section h2 {
font-size: 1.3rem;
margin-bottom: 1rem;
}

.review-card {
background: var(--card);
border-radius: var(--radius);
box-shadow: var(--shadow);
padding: 1.25rem 1.5rem;
margin-bottom: 1rem;
border-left: 4px solid var(--border);
}
.card-correct { border-left-color: var(--success); }
.card-wrong   { border-left-color: var(--danger); }
.card-skipped { border-left-color: var(--warn); }

.review-header {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 0.6rem;
}

.review-qnum {
font-size: 0.75rem;
font-weight: 700;
letter-spacing: 1px;
color: var(--muted);
text-transform: uppercase;
}

.review-status { font-size: 0.85rem; font-weight: 600; }

.review-question {
font-size: 1rem;
font-weight: 600;
margin-bottom: 0.75rem;
line-height: 1.4;
}

.review-options {
display: flex;
flex-direction: column;
```

```
gap: 0.4rem;
}

.review-option {
font-size: 0.9rem;
padding: 0.45rem 0.75rem;
border-radius: 8px;
color: var(--muted);
}
.opt-correct { background: #D5F5E3; color: #1D8348; font-weight: 600; }
.opt-wrong   { background: #FADBD8; color: #922B21; font-weight: 600; }

.review-note {
font-size: 0.85rem;
color: var(--warn);
margin-top: 0.5rem;
}

/* ========== RESPONSIVE ========== */
@media (max-width: 540px) {
.info-grid   { grid-template-columns: repeat(2, 1fr); }
.score-stats { grid-template-columns: repeat(2, 1fr); }
.hero-card   { padding: 2rem 1.25rem; }
.extra-stats { flex-direction: column; gap: 0.5rem; }
}
```

❖ Output


Quiz Master
Web Programming Lab
Welcome to Quiz Master
Test your knowledge in Web Development, Python & Flask
10 Questions
5 Min Timer
Shuffled
-0.25 Marking
Rules
Each correct answer earns +1 mark
Each wrong answer deducts -0.25 marks
Skipped questions carry no penalty
Questions and options are shuffled every attempt
Timer auto-submits at 5 minutes
Start Quiz


Quiz Master
04:54
0 / 10 answered
Q1
Not answered
Which Python framework is used for building web applications in this course?
Django
Flask
Tornado
FastAPI
Q2
Not answered
What does API stand for?
Automated Processing Integration
Application Process Input
Application Programming Interface
Advanced Program Index


Quiz Master
Results
F
Quiz Completed!
Don't give up! Practice makes perfect.
0
CORRECT
0
WRONG
10
SKIPPED
0.00 / 10
FINAL SCORE
Time Taken: 0m 23s
Penalty: -0.00
Restart Quiz
Home
Answer Review
Q1
Skipped (0)
Which Python framework is used for building web applications in this course?
Django
Flask
Tornado
FastAPI
You skipped this question. Correct answer: Flask